DSCI 560 - Lab Assignment 7
Mobile Phone-based IoT Network with IoT Hub in Cloud
Instructor: Young Cho, Ph.D.

1. Turn Mobile Phone into Internet of Things

Go to https://owntracks.org to set up your phone. Download and install the Owntracks App for your iPhone and/or Android phone.

Go through the guide https://owntracks.org/booklet/, including the Quick setup, to understand what Owntracks can do while gaining hands-on experience with IoT and cloud infrastructure that work with the IoT.

Once you've gained some experience, you can go to the next section, where you will set up an IoT hub on AWS for your Owntracks App.

2. Set up Thingsboard Community Edition on Amazon Web Services

Configure and launch an Ubuntu Linux EC2 instance. A free-tier instance does work, but it is awfully slow to set up and run. So, you may want to use a larger instance with a modest fee. Make sure to use/configure your security group to allow all traffic from anywhere IPv4 (0.0.0.0/0). Moreover, make sure to create and download your security key. The downloaded file will have a name with ".pem" or ".ppk" at the end.

Then, use the "ssh" terminal program to remotely log into the EC2 node.
https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/connect-linux-inst-ssh.html

Your username should be ubuntu and if you are using Linux or Mac terminal with OpenSSH, you should be able to log in by typing the following (if your key file name is "key.pem" and your public IP address is "52.87.235.210"):

```
ssh -i key.pem ubuntu@52.87.235.210
```

If you need to brush up on the Linux command line interface, please go through the following tutorial.
https://ubuntu.com/tutorials/command-line-for-beginners#1-overview

Now, you can download and install Thingsboard Community Edition onto this EC2 Instance

Step 1. Install Java 17 (OpenJDK)

ThingsBoard service is running on Java 17. To install OpenJDK 17, follow these instructions

```
sudo apt update && sudo apt install openjdk-17-jdk
```

Configure your operating system to use OpenJDK 17 by default. You can configure the default version by running the following command:

```
sudo update-alternatives --config java
```

To check the installed Java version on your system, use the following command:

```
java -version
```

The expected result is:

```
openjdk version "17.x.xx"
OpenJDK Runtime Environment (...)
OpenJDK 64-Bit Server VM (...)
```

Step 2. ThingsBoard service installation

Follow the instructions on https://thingsboard.io/ to download the install FREE community edition

Step 3. Configure ThingsBoard database

ThingsBoard can be configured to work with variety of database, including MySQL and PostgresSQL. See the corresponding architecture page for more details.

PostgreSQL Installation

To install the PostgreSQL database, run these commands:

```
# Automated repository configuration:
sudo apt install -y postgresql-common
sudo /usr/share/postgresql-common/pgdg/apt.postgresql.org.sh

# install and launch the postgresql service:
sudo apt update
sudo apt -y install postgresql-16
sudo service postgresql start
```

Once PostgreSQL is installed, it is recommended to set the password for the PostgreSQL main user.

The following command will switch the current user to the PostgreSQL user and set the password directly in PostgreSQL.

```
sudo -u postgres psql -c "\password"
```

Then, enter and confirm the password.

Then, connect to the “postgres” database as the “postgres” user:

```
psql -U postgres -d postgres -h 127.0.0.1 -W
```

Within psql prompt, type the following to create a ThingsBoard database named “thingsboard” :

```
CREATE DATABASE thingsboard;
exit
```

Then edit ThingsBoard configuration file

```
sudo nano /etc/thingsboard/conf/thingsboard.conf
```

Add the following lines to the configuration file. Don’t forget to replace “PUT_YOUR_POSTGRESQL_PASSWORD_HERE” with your PostgreSQL user password:

```
# DB Configuration
export DATABASE_TS_TYPE=sql
export SPRING_DATASOURCE_URL=jdbc:postgresql://localhost:5432/thingsboard
export SPRING_DATASOURCE_USERNAME=postgres
export SPRING_DATASOURCE_PASSWORD=PUT_YOUR_POSTGRESQL_PASSWORD_HERE
```

```
# Specify partitioning size for timestamp key-value storage. Allowed
values: DAYS, MONTHS, YEARS, INDEFINITE.
export SQL_POSTGRES_TS_KV_PARTITIONING=MONTHS
```

Step 4. Choose ThingsBoard queue service

ThingsBoard can use various messaging systems/brokers for storing the messages and communication between ThingsBoard services.

In Memory queue implementation is built-in and default. It is useful for development environments and is not suitable for production deployments or any sort of cluster deployments.

There are other options for various needs. You may want to explore them later.

Step 5. Run the installation script

Once ThingsBoard service is installed and DB configuration is updated, you can execute the following script:

```
sudo /usr/share/thingsboard/bin/install/install.sh --loadDemo
```

This step can take a while if you are using a free-tier instance. If you increase its memory for a charged version of an instance, the installation can go faster.

Step 6. Start ThingsBoard service

Execute the following command to start ThingsBoard:

```
sudo service thingsboard start
```

Now, you should be able to access the ThingsBoard User Interface using a web browser to go to the following web address (a given public IP address of 52.87.235.210).

```
http://52.87.235.210:8080/
```

If you are unable to get access to your own ThingBoard web interface on your Ubuntu instance, you may have to modify your security group setup. These settings can be tricky and may prevent external users from accessing the installed ThingsBoard on EC2. One definite way to ensure access is to open all connections from all IP addresses and ports.

You can modify the security settings of the group by going to the Security group. Then click on the specific security group for your instance to “Edit inbound rules”. There, you can “Add rule” where you can specify the type to be “All traffic” and the Source to be Anywhere IPv4. Then save the rules. Once this modification is made, you can reboot your instance.

3. Configurinh ThingsBoard

You may want to review the following guide to learn to use ThingsBoard.

https://thingsboard.io/docs/guides/

ThingsBoard is a platform that is like many commercial IoT hubs on the cloud, which often has fees. Therefore, if you learn how to use this platform, other commercial IoT hubs will be easier to learn and use in a relatively short time.

The following default credentials are available if you have specified –loadDemo during execution of the installation script:

```
System Administrator: sysadmin@thingsboard.org / sysadmin
Tenant Administrator: tenant@thingsboard.org / tenant
Customer User: customer@thingsboard.org / customer
```

Log in as a tenant@thingsboard.org with the password "tenant". You can always change passwords for each account on the account profile page.

Now, go to create a device for your phone by going to Entities->Devices, then select +, then Add new device. Fill in the Name and use the "default profile" then Add. Then it should pop up a window showing connectivity for HTTP, MQTT, and CoAP for different Operating Systems. Write down only the URL (ignore curl and other options) under "Execute the following command":

Example: `http://52.87.235.210:8080/api/v1/oylNvRDLFeTJfpszfA56/telemetry`

Then Click on Close to set up your device on ThingBoard. For the example, note that `oylNvRDLFeTJfpszfA56` is the device token.

4. Setting up Owntracks App to Send HTTP messages to ThingsBoard

Download and install an App named owntracks from Apple or Google App store.

On the App, go to Preferences

If everything was entered correctly, your phone should be sending several information to ThingsboardIO every 1 second.

5. IoT Hub Dashboard

To verify that the phone data is being sent to Thingsboard, select your Device under Entities – Devices. Select the Latest telemetry tab under Device detail. If all is well, then you will see the Values changing over time. You may have to click on the arrow at the bottom right of the window to see the longitude and latitude values changing.

Now, you can visualize this on Thingsboard in real-time by creating a dashboard.

Go to Dashboards and click on + to create a new dashboard for your devices. Type in the title, then Add. Now, click on Add new widget to see different widgets that you can place in your dashboard. You can experiment with different widgets to see what they can do later. For now, click on the Maps, then OpenStreet Map.

Pull down the type to select Device. Then select your device name that you assigned to the Phone. For the Data keys, type "lat" to select latitude then "lon" to select longitude. Then select the Appearance tab right next to the Data tab. Then go to where it says Latitude key name and change it to lat, and Longitude key name to lon. Then click on Apply.

If all goes well, then you should see your position on the map. If you move your phone around, you should also see the updated movement in the map every second.

6. Custom Mapping Program

Now, you should have a pretty good idea of how these things work. Given these new tools, you ought to be able to make custom end-to-end IoT solutions. For this lab, try to create some sort of interesting mapping solution using multiple phone data. You can use GPS data as well as other additional sensor data that you may have access to from owntracks app.

Submission

Provide a video per team that demonstrates the entire working solution. The video should demonstrate that you have completed the assignment tasks, as well as demonstrate your final application and discuss the design decisions and reasoning. Also, include details about how your team worked together on this project. Please include the team's name and the names of all team members in the video.

NOTE: Once all is done, make sure to stop/terminate the EC2 instance and properly exit the Owntracks App. Otherwise, you may get charged for the cloud instance, and/or your phone battery will drain quickly as the app will run in the background.

There will be a 50% penalty for all late submissions.